Dan Bachar

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Software engineer with 10+ years of production experience across distributed systems, backend, and full-stack. Currently completing an M.Sc. at TU Munich (Distributed Systems, Computer Networks & Architecture) with a joint thesis with the Technion on peer-to-peer messaging protocols. Previously led applied research at Siemens on distributed industrial IoT and 6G infrastructure for EU-funded projects. Earlier job experience spans health, industrial IoT, mission-critical government systems, and leading small teams

EXPERIENCE

Master's Thesis · Connected Mobility Chair, TU Munich & Technion, Haifa *2025 – 2026*

- Designing and implementing a secure, offline-first, heterogeneous peer-to-peer messaging protocol as the communication backbone for a distributed social-graph based ecosystem, including digital contracts, digital assembly, currency
- Joint research collaboration between Technion, TU Munich, and the Weizmann Institute

Researcher & Distributed Systems Developer · Siemens AG, Munich *2023 – 2025*

- Led R&D of demonstrations for distributed industrial IoT edge systems as part of the EU-funded Edgeless research project
- Designed a graph-based configuration tool for edge network topology
- Built a cluster management platform for industrial computers with system-level snapshotting for research demos
- Build a 6G-based satellite communication handover system for next-generation EU rail infrastructure

Frontend Tech Lead · Applied Security GmbH / SITS AG, Großwallstadt *2022 – 2023*

- Led frontend development of the public statutory health insurance portal for a major German provider (DASBV GmbH)
- Introduced state management architecture, clean-code standards, TDD and static analysis; increased test coverage to 80%

Fullstack Developer · mantro GmbH, Munich *2018 – 2019*

- Employee mobility platform (Mobiko): designed an algorithmic solution to a variant of the knapsack problem for tax-optimized business expense allocation
- Hunting application (Jagdgefährte): optimized database query performance, reducing request latency by 25%
- Industrial IoT / computer vision device (company42): led backend development and product ownership; designed a scalable parallelized inference layer for querying edge-deployed CV models

Frontend Engineer Cellebrite, Petah Tikva *2017*

- Product ownership and frontend development of a smartphone diagnostics and repair platform
- Cross-team integration lead across backend, mobile, customization, and core modules
- Lead client demonstrations, restructured customer feedback loops to integrate overseas customers into development sprints

Fullstack Developer · Matzpen, C4I Corps, IDF *2014 – 2017*

- Designed and maintained a cross-network database synchronization system between civilian and military networks for the IDF's public web infrastructure
- Maintained mission-critical enterprise applications supporting military systems
- Led the modernization of a legacy monolithic codebase, migrating JSP web apps to modern component-based frontend using Angular

SKILLS

Languages:	Rust, Go, Python, Java, C#, C, TypeScript / JavaScript, Node.js, Dart, SQL
Frameworks & Runtimes:	NestJS, React, Angular, Protractor, Svelte, Flutter, Redux, Tokio, Axum
Systems & Infrastructure:	Kubernetes, Docker, Redis, Linux, CI/CD pipelines, GCP, Ansible, QEMU, PostgreSQL, Supabase
Communication Protocols:	MQTT, gRPC, WebSockets, GraphQL, UDP, BLE, 6LoWPAN, WiFi, HTTPS, TCP, UDP
Embedded / Hardware:	Arduino, Raspberry Pi, ESP32
AI-assisted development:	Claude, Codex, Copilot

EDUCATION

Technical University of Munich — M.Sc. Informatik · *2023 – 2026*
Major: Distributed Systems, Computer Networks & Architecture

Technical University of Munich — B.Sc. Informatik · *2019 – 2023*

LANGUAGES

Hebrew - native **English** — bilingual **German** — C1